

MAMATHA VELTURI

+1 7373827247 | velturimamatha29@gmail.com | [LinkedIn](#)

PROFESSIONAL SUMMARY

Senior Data & AI Engineer with 10+ years of experience designing and deploying production-grade data and AI solutions. Skilled in architecting end-to-end ML pipelines on AWS SageMaker and Amazon Bedrock, integrating RAG architectures and multi-agent systems to automate decision workflows. Proven ability to deliver scalable AI-driven systems that drive operational efficiency and compliance in regulated environments.

TECHNICAL SKILLS

- **AI / ML / GenAI:** LLM Integration, RAG Architecture, Generative AI, NLP, Prompt Engineering & Chaining, Agentic AI, Multi-Agent Systems, Model Evaluation & Benchmarking, Machine Learning, Large Language Models, Conversational AI, Customer AI Agents
- **ML Frameworks:** LangChain, LangGraph, Semantic Kernel, Amazon Bedrock (Foundation Models, Guardrails, Knowledge Bases), SageMaker, MLflow
- **Vector Databases:** FAISS, Pinecone, Amazon OpenSearch, pgvector - semantic search, embeddings, RAG indexing
- **Languages:** Python (LangChain, Boto3, NumPy, Pandas, Scikit-learn), SQL, Shell Scripting, REST API Integration
- **Cloud - AWS:** SageMaker, Bedrock, Glue, Redshift, S3, Lambda, Step Functions, Kinesis, API Gateway, ECS/ECR, CloudWatch, IAM, Athena, CloudFormation, Amazon Connect, Amazon Lex, EventBridge
- **Cloud - Azure / GCP:** Azure Data Factory, Azure Data Fundamentals, GCP Familiarity, Azure OpenAI, Azure AI Services, Azure Event Hubs
- **Data Platforms:** Snowflake (SnowPipe, Tasks, Streams, Time Travel, Optimizer), Databricks (PySpark, Delta Lake, Unity Catalog, MLflow), dbt
- **ETL / Pipeline Tools:** AWS Glue, Informatica IDMC / IICS / PowerCenter / IDQ, SSIS, ADF, Apache Airflow
- **Pipeline & Data Processing:** Rag Pipeline, Data Scraping, Data Ingestion, Data Chunking, Data Preparation
- **MLOps:** MLflow, Databricks MLflow, Kubeflow (familiar), model versioning, A/B testing, CI/CD for ML, Docker, Terraform, CloudFormation, Full Stack AI Deployment, MCP Servers, Logging
- **Databases:** Amazon Redshift, Snowflake, MS SQL Server, PostgreSQL, DynamoDB (NoSQL), Teradata
- **Data Architecture:** Kimball Modeling, Star/Snowflake Schema, OLTP/OLAP, Data Lake, Lakehouse, Data Lineage, Unity Catalog, Glue Data Catalog
- **Compliance / Security:** HIPAA, GDPR, AWS IAM, RBAC, Data Masking, Bedrock Guardrails, GxP Compliance, EU Annex 11, FDA 21 CFR Part 11, Pharmaceutical Industry Compliance
- **Reporting / BI:** Power BI, Tableau, SSRS, CloudWatch Dashboards
- **Version Control / IaC:** Git, Terraform, AWS CloudFormation, Ansible, YAML
- **Streaming & Event-Driven Architectures:** Real-time Data Streaming, Event-driven Architectures, Kafka, Azure Event Hubs
- **AI Governance & Ethics:** AI Safety, Responsible AI, Explainability Practices

CERTIFICATIONS

- AWS Solutions Architect - Associate: 2025
- Databricks Certified - Databricks Fundamentals: 2025
- Microsoft Certified - Azure Data Fundamentals: 2024

PROFESSIONAL EXPERIENCE

Capital One

Oct 2024 - Present

Senior Data & AI Engineer

- Architected and deployed end-to-end ML pipelines on AWS SageMaker for auto loan risk scoring, integrating feature engineering, model training, hyperparameter tuning, and automated retraining triggers to support real-time decisioning across the Auto Finance platform.
- Built production-grade RAG (Retrieval-Augmented Generation) pipelines using Amazon Bedrock Knowledge Bases, FAISS, and Pinecone vector databases - enabling accurate, grounded LLM responses for internal Q&A and analyst tools, reducing manual research time by 45%.
- Designed and deployed autonomous multi-agent AI systems using LangChain, Langgraph, and Amazon Bedrock, coordinating specialized sub-agents for document retrieval, summarization, and clinical-style decision-making; reduced manual analyst effort by 45% across operations teams.
- Integrated Amazon Bedrock Guardrails to enforce content safety, policy compliance, and hallucination reduction across all AI agent outputs, applying responsible AI and explainability practices aligned with HIPAA-compliant deployment standards.
- Evaluated and benchmarked foundation models (Claude, Titan, Llama 2) on Amazon Bedrock for task-specific accuracy, latency, and cost trade-offs; optimized prompts for conversational AI virtual assistants and internal agents
- Deployed AI-powered REST APIs using AWS Lambda and API Gateway to expose Bedrock and SageMaker inference endpoints, enabling seamless downstream integration with business dashboards and operational applications.

- Engineered MLOps workflows using MLflow on Databricks for experiment tracking, model registry, and versioning - establishing a reproducible model lifecycle process across the data science team.
- Developed scalable ETL/ELT pipelines in Databricks (PySpark) and AWS Glue to process multi-terabyte structured and semi-structured datasets, including data scraping, ingestion, chunking, and preparation, feeding clean, governed data into ML feature stores and SageMaker training jobs.
- Built NLP-based text classification and entity extraction models using Python and Hugging Face transformers to automate processing of unstructured loan application documents, cutting review time by 30%.
- Implemented real-time data ingestion using Amazon Kinesis Data Streams and AWS API Gateway to capture loan application status updates for downstream ML model inference pipelines.
- Implemented real-time data ingestion using Amazon Kinesis Data Streams, Kafka, and Azure Event Hubs to capture loan application status updates for downstream ML model inference pipelines within event-driven architectures.
- Built and maintained dbt transformation pipelines on Snowflake to standardize feature data models, improve data quality, and accelerate ML training data preparation cycles for cross-functional analytics teams.
- Integrated Amazon Connect with AI-powered conversational interfaces, developing Customer AI Agents using Amazon Lex and Amazon Bedrock to enhance customer engagement workflows
- Configured EventBridge rules to orchestrate real-time events between Amazon Connect and backend Lambda functions, enabling seamless AI-driven contact center interactions
- Implemented data lineage and governance using Unity Catalog and AWS Glue Data Catalog, ensuring audit readiness and reproducibility for all ML pipeline inputs and outputs.
- Led full stack AI deployment initiatives, managing end-to-end development, deployment, and monitoring of generative AI solutions across cloud and on-premise environments.
- Implemented event-driven architectures using Amazon EventBridge and Kafka to power real-time data streaming and autonomous ML-driven workflows, improving system responsiveness and scalability.

Merck

Jan 2021 - Nov 2023

Senior ETL Data Engineer

Remote

- Designed and deployed GxP-compliant LLM-powered document retrieval workflows at Merck using LangChain prompt chaining, Amazon Bedrock APIs, and vector search (Pinecone, FAISS) in adherence with FDA 21 CFR Part 11 and EU Annex 11, improving answer relevance for scientific literature Q&A by 40%.
- Automated Bedrock model inference pipelines using AWS Lambda and Step Functions, integrating post-processing logic to route LLM outputs to downstream consumers and clinical reporting tools via REST API integration.
- Built and maintained ML data pipelines on Databricks (PySpark) and Snowflake for pharmaceutical research analytics, processing large structured and semi-structured datasets to feed model training and reporting workflows.
- Architected Snowflake-based data warehouses on AWS integrating OLTP and OLAP sources, enabling real-time analytics and serving as the governed data foundation for AI model feature stores.
- Implemented vector database indexing using Pinecone and FAISS to store and retrieve document embeddings for LLM powered search and retrieval, improving AI search relevance for scientific content by 40%.
- Designed a data lake solution on Amazon S3 with Amazon OpenSearch as a semantic search layer, laying the RAG pipeline foundation for AI-powered research applications.
- Engineered Python data quality automation scripts to monitor and validate ML pipeline inputs across enterprise systems, reducing data-related model failures by 40% and improving overall model reliability.
- Led cloud migration of on-premises pharmaceutical data to AWS using Informatica IDMC, ensuring data integrity and minimal downtime for mission-critical research data pipelines.
- Coordinated Databricks cluster provisioning via Terraform, implementing auto-scaling and cluster termination policies that reduced cloud compute costs by 20%.
- Developed and optimized complex SQL procedures, Snowflake SnowPipe ingestion, Tasks, and Time Travel workflows to support enterprise ML data processing and audit requirements.
- Collaborated with analytics and BI teams to deliver on-demand reporting solutions in Power BI and Tableau, reducing manual data analysis effort by 50% across global research teams.

Seaspan

Jun 2019 - Oct 2019

Senior ETL / Data Engineer

Vancouver, BC, Canada

- Designed and implemented large-scale Snowflake Cloud Data Warehouse solutions on AWS, supporting batch and real time analytics workloads with optimized query performance and cost-efficient clustering.
- Integrated Databricks with AWS S3, EC2, RDS, IAM, and Glue to build secure, scalable data pipelines for both batch and streaming data ingestion needs.
- Developed Python automation scripts to orchestrate AWS Glue and Lambda workflows, reducing manual data processing overhead by 40% and enabling self-service analytics.
- Improved data retrieval speed by 25% through Glue Catalog integration and optimized S3 data partitioning and storage policies.
- Implemented data governance protocols including IAM RBAC and data masking to ensure data security, privacy, and

HIPAA compliance across all pipelines.

TransLink - Metro Vancouver

ETL Data Engineer

Canada

Jan 2016 - Mar 2019

Vancouver, BC,

- Leveraged Informatica IICS and PowerCenter to design complex ETL workflows for cloud and hybrid environments, implementing advanced data transformation and cleansing techniques.
- Built Python ETL scripts to automate extraction and transformation of transit operational data, reducing report generation cycle from 3 days to 4 hours for cross-functional business teams.
- Led data migration projects using Informatica and AWS services, transferring large datasets securely from legacy databases to cloud-native architectures.
- Designed and optimized Informatica PowerCenter mappings and complex SQL (DDL/DML, CTEs, stored procedures, triggers) to support enterprise reporting and operational analytics.
- Integrated S3, EC2, RDS, and IAM within Databricks environments to build secure, scalable data lake solutions; improved data retrieval speed by 25% via Glue Catalog optimization.
- Conducted regular HIPAA compliance audits across all data pipelines, ensuring sensitive data was appropriately protected in accordance

AXA Canada Inc.

ETL Developer

Sep 2013 - Oct 2015

Montreal, QC, Canada

- Developed SSIS packages for enterprise ETL workflows, implementing complex data transformation, cleansing, and loading processes for financial reporting systems.
- Wrote complex T-SQL queries, stored procedures, and SSRS reports for business intelligence and ad hoc analytics across insurance data domains.
- Managed SQL Server environments (2008–2014) including installation, configuration, index optimization, and performance tuning to support high-volume transactional workloads.

Nico Info Systems India

SQL Server Developer

Jul 2011 - Aug 2013

India

- Designed and developed SSIS ETL packages for data migration, cleansing, and validation; created source-to-target mappings and functional specifications aligned with business requirements.
- Built SSRS tabular and matrix reports; developed stored procedures, views, triggers, and indexes in MS SQL Server to support operational data needs.

EDUCATION

JNTU, Hyderabad

Master of Technology

2009 - 2011

Hyderabad

JNTU, Hyderabad

Bachelor of Technology

2005 - 2009

Hyderabad